(Please write your enrollment number immediately)

Mid-Term Examination

B.Tech Programmes (Under the Aegis of USICT)
6th Semester, (March 2025)

Paper Code: CIE 310T **Subject: Advance DBMS**

Time: **1.5 Hours** Max Marks: **30**

Note: Attempt Q.1, which is compulsory and any two more questions from the remaining.

Q.No	Question	Max Marks	CO
Q1	a) Give any 2 significant impacts of emerging database standards on distributed database management systems (DDBMS). b) Discuss the role of a Storage Manager in a DBMS. c) Why are key constraints important in a distributed database management system? d) Explain any one query optimization challenge in a distributed database. e) What is a database trigger and state any one common use case for triggers?	2*5 =10	CO1 CO1 CO2 CO2 CO1
Q2	a) Differentiate between Embedded SQL and Dynamic SQL with examples. b) Explain the concept of Data Replication and its significance in distributed databases.	5 5	CO1 CO2
Q3	a)How does Parallel Database Architecture improve query performance? Explain with a diagram. b) A company needs to ensure secure database transactions. Propose a solution that involves security considerations and integrity constraints.	5 5	CO1 CO2
Q4	a) What are ACID properties in database transaction processing? Explain their importance with an example. b) Consider a relational database with **Orders (Order_ID, Customer_ID, Amount)** and **Customers (Customer_ID, Name, Address)**. Make sample tables on your own. **i.** Write an SQL query to retrieve the names of customers who have placed orders of more than $5000. **ii.** Count the number of orders placed by each customer, displaying Customer Name and Order Count.	5 5	CO1 CO2

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SUPPLEMENTARY MID TERM EXAMINATION

B.Tech Programmes (Under the Aegis of USICT)

IV Semester, March 2025

Paper Code: CIC-210 **Subject: Database Management System**

Time: 1.5 Hrs **Max Marks: 30**

Note: Attempt Question 1, which is compulsory, and any two more questions from the remaining.

Q. No.		Question	Max Marks	CO(s)
1.	a)	Explain the difference between Physical data independence and Logical data independence.	2.5	CO1
	b)	Who is a Database Administrator? Explain the responsibilities of a database administrator (DBA).	2.5	CO1
	c)	Explain the concept of aggregation with an example.	2.5	CO1
	d)	What are the different types of database languages? Explain with examples.	2.5	CO2
2.	a)	Define the terms entity, attribute, and relationship with suitable examples.	5	CO1
	b)	Draw an ER diagram for a Library Management System, considering entities like Books, Members, Librarians, and Transactions. Clearly indicate attributes and relationships.	5	CO1
3.		Consider the following relational schema: Customers (Customer_ID, Name, Address, Phone) Accounts (Account_No, Customer_ID, Account_Type, Balance) Transactions (Transaction_ID, Account_No, Transaction_Type, Amount, Date) Write SQL Command and Relational algebra expression for each of following queries. i) Find the names and account numbers of customers who have a savings account. ii) Retrieve the total balance of all accounts in the bank. iii) Find customers who have a balance of more than ₹50,000. iv) Find the total balance for each account type. v) Find account types where the total balance exceeds ₹1,00,000.	2*5=10	CO2
4.	a)	Differentiate between Primary Key and Foreign Key with an example.	5	CO2
	b)	Explain Left Outer Join, Right Outer Join and Full Outer Join with an example.	5	CO2

(Please write your Exam Roll No.) *Exam Roll No.....*

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

FOURTH SEMESTER - MARCH, 2025

Paper Code: CIC-210 **Subject: Database Management Systems**

Time: 1 ½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No.1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO(s)
Q-1 (a)	We can convert any weak entity set to a strong entity set by simply adding appropriate attributes. Why, then, do we have weak entity sets?	2.5	CO1
(b)	What is data independence in DBMS? Explain logical and physical data independence with the help of a diagram.	2.5	CO1
(c)	Explain the concept of Aggregation in E-R diagrams with the help of an example.	2.5	CO1
(d)	Explain the Division operation in relational algebra with the help of an example.	2.5	CO2
Q-2 (a)	Differentiate between Super key, Candidate key and Primary key with an example.	4	CO1
(b)	Give an example of n-ary relationship (where n=3 and n=4). Also create relations from this relationship.	4	CO1
(c)	How do you map a relationship set to a table in a database?	2	CO1
Q-3(a)	Construct an E-R diagram for a vehicle insurance company whose customers have one or more vehicles. Each vehicle has associated with it zero to any number of recorded accidents.	3	CO1
(b)	Let the following relation schemas be given: R= (A, B, C) S = (D, E, F) Let relations r (R) and s(S) be given. Give an expression in SQL that is equivalent to each of the following queries. (a) $\sigma_{B=17}$ (r) (b) $\Pi_{A,F}$ ($\sigma_{C=D}$ (rxs))	4	CO2
(c)	What is natural join? How do you represent this in relational algebra? Give an example to express this join.	3	CO2
Q-4 (a)	Consider the following relational schema: Sailors (sid, sname, rating, age) Boats (bid, name, color) Reserves (sid, bid, day) Write SQL or Relational algebra expression for each of the following queries: (i) Find the age of the sailors who have ratings of atleast 10. (ii) Find the names of sailors who have reserved boat with id no 27. (iii) Delete all tuples in the boats relation where color of the boat is yellow. (iv) Increase the rating of all sailors by 2%.	8	CO2
(b)	What is the difference between DDL and DML?	2	CO2

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(Please write your Enrolment no. immediately) Enrolment No._____

SUPPLEMENTARY MID TERM EXAMINATION

B.Tech Programmes (Under the Aegis of USICT)

IV Semester, March 2025

Paper Code: CIC-210 **Subject: Database Management System**

Time: 1.5 Hrs **Max Marks: 30**

Note: Attempt Question 1, which is compulsory, and any two more questions from the remaining.

Q. No.		Question	Max Marks	CO(s)
1.	a)	Explain the difference between Physical data independence and Logical data independence.	2.5	CO1
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4.	a)	Differentiate between Primary Key and Foreign Key with an example.	5	CO2
	b)	Explain Left Outer Join, Right Outer Join and Full Outer Join with an example.	5	CO2

(Please write your Exam Roll No.) *Exam Roll No.....*

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

FOURTH SEMESTER - MARCH, 2025

Paper Code: CIC-210 **Subject: Database Management Systems**

Time: 1 ½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No.1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO(s)
Q-1 (a)	We can convert any weak entity set to a strong entity set by simply adding appropriate attributes. Why, then, do we have weak entity sets?	2.5	CO1
(b)	What is data independence in DBMS? Explain logical and physical data independence with the help of a diagram.	2.5	CO1
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(d)	Explain the Division operation in relational algebra with the help of an example.	2.5	CO2
Q-2 (a)	Differentiate between Super key, Candidate key and Primary key with an example.	4	CO1
(b)	Give an example of n-ary relationship (where n=3 and n=4). Also create relations from this relationship.	4	CO1
(c)	How do you map a relationship set to a table in a database?	2	CO1
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(b)	Let the following relation schemas be given: R= (A, B, C) S = (D, E, F) Let relations r (R) and s(S) be given. Give an expression in SQL that is equivalent to each of the following queries. (a) $\sigma_{B=17}$ (r) (b) $\Pi_{A,F}$ ($\sigma_{C=D}$ (rxs))	4	CO2
(c)	What is natural join? How do you represent this in relational algebra? Give an example to express this join.	3	CO2
Q-4 (a)	Consider the following relational schema: Sailors (sid, sname, rating, age) Boats (bid, name, color) Reserves (sid, bid, day) Write SQL or Relational algebra expression for each of the following queries: (i) Find the age of the sailors who have ratings of atleast 10. (ii) Find the names of sailors who have reserved boat with id no 27. (iii) Delete all tuples in the boats relation where color of the boat is yellow. (iv) Increase the rating of all sailors by 2%.	8	CO2
(b)	What is the difference between DDL and DML?	2	CO2

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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

IV Semester, April, 2024

Paper Code: CIC 210 **Subject: Database Management System**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	a. Differentiate between a strong and weak entity type with an example of each.	2.5	CO1
	b. Differentiate between physical and logical data independence.	2.5	CO1
	c. Explain DDL, DML and DCL in SQL. List the commands under each category.	2.5	CO2
	d. List different types of attributes of an entity with an appropriate example.	2.5	CO1
2	a. Construct an ER diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	5	CO1
	b. Explain the following concepts w.r.t Extended ER diagrams (with an appropriate Example): i) Specialization ii) Aggregation	5	CO1
3	Consider the following relational schema: *employee (employee name, street, city, manager name)* *works (employee name, company name, salary)* *company (company name, city)* write SQL or Relational algebra expression for each of the following queries. i) Find the names of all employees who live in the same city as the company for which they work. ii) Find the names and cities of residence of all employees who work for Small Bank Corporation. iii) Find the total number of employees working in each company. iv) List the name of the employees who are working in the same company where the employee Smith is working. (Note: Smith is the name of an employee) v) Name the employee who is earning maximum salary.	2*5=10	CO2
4.	a. Differentiate between Super Key, Candidate Key and Primary Key with an example.	5	CO2
	b. Explain natural join, left outer join, right outer join w.r.t SQL with example.	5	CO2

(Please writ ur Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-210 Subject: **Database Management System**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt questions in all including Q.No.1 which is compulsory. Select one question from each Unit.

Q1 a) Give an example of mapping of generalization or specialization into relation schemas Summarize the steps involved in converting the ER constructs to relational schema. **(3)**

b) What is a Weak Entity type? Explain the role of partial key in design of weak entity type. **(3)**

c) Consider a relational schema with the following tables:
Employee (EmpID, EmpName, DeptID)
Department (DeptID, DeptName)
Write a relational calculus expression to retrieve the names of all employees who work in the "Sales" department. **(3)**

d) Why is the domain key normal form (DKNF) known as ultimate normal form? **(3)**

e) Compare and contrast heap files and sorted files in terms of storage efficiency and query performance. When would you choose to use a heap file over a sorted file, and vice versa? **(3)**

UNIT-I

Q2 a) Design an ER — Diagram for a UNIVERSITY database schema taking into account at least five entities and indicate all keys and cardinality constraints and assumptions that are made. **(7)**

b) Define the following with examples: **(5)**
(i) Super key (ii) Candidate key
(iii) Primary key (iv) Foreign key

c) What do you mean by constraints? Differentiate between table and column constraint. Explain with an example. **(3)**

Q3 a) Consider the following tables: **(8)**
works (Pname, Cname, Salary)
lives (Pname, Street, City)
located-In (Cname, City)
write the following queries in SQL:
i) List the names of the people who work for the company 'Wipro' along with the cities they live in.
ii) Find the names of the persons who do not work for 'Infosys'.
iii) Find the people whose salaries are more than that of all of the 'oracle' employees.
iv) Find the persons who works and lives in the same city.

b) Differentiate between DDL, DML, and DCL in SQL. Provide examples for each category and explain their roles in database management. **(4)**

c) Explain the various inner join operations in relational algebra with examples. **(3)**

P.T.O.